

Adaptive Interval Calibration for DeepPIPE under Distribution Shift: Two-Phase Training and State-Dependent Conformal Control on the NIFTY 50

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Abstract—Distribution-free prediction intervals are attractive for financial forecasting because they do not assume a parametric error model, yet their reliability under distribution shift remains poorly understood. We study DeepPIPE—an encoder-decoder LSTM with three output heads trained by a hybrid point-width-coverage loss—on 1,676 daily NIFTY 50 observations (2020–2026), where the test regime lies predominantly above the maximum training price (24,613 vs. 26,329), creating a severe out-of-distribution (OOD) setting. Single-phase training collapses: mean prediction interval coverage probability (PICP) is only 0.32 ± 0.27 across five seeds, with one seed reaching zero coverage. A two-phase procedure—point-only warm-up followed by full hybrid loss—restores calibration to 0.99 ± 0.01 without architectural change. Even after this fix, three limitations persist: (G1) volatility-blind widths (width-VIX correlation -0.50); (G2) coverage collapse under rolling-window regime breaks (PICP 0.01); and (G3) over-coverage insensitive to the requested level (≈ 0.995 at 80/90/95%). We introduce DeepPIPE-Ctrl, an additive contextual-bandit controller that replaces the static split-conformal half-width $\hat{q}_\alpha$ with a state-dependent multiplier $m_t \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$, trained offline to minimise the Winkler interval score. On held-out data, conformal PID attains the closest target coverage (PICP 0.901) and partially corrects the VIX-width correlation; learned linear and neural policies do not yet dominate static conformal, revealing a data-limited but informative negative result. We release a reproducible open-source implementation with walk-forward evaluation, statistical tests, and width-gated economic backtests.

Index Terms—prediction intervals, distribution shift, DeepPIPE, conformal prediction, contextual bandits, NIFTY 50, uncertainty quantification.

I. Introduction

Point forecasts alone are insufficient for portfolio construction, risk budgeting, and position sizing. Prediction intervals (PIs) augment a point estimate $\hat{y}_{t+1}$ with bounds $[L_{t+1}, U_{t+1}]$ intended to contain the realised close y_{t+1} with probability p . In finance, PIs must remain valid when market regimes drift: training and test distributions rarely coincide.

DeepPIPE [1] is a distribution-free PI framework based on an encoder-decoder LSTM with three heads (point, lower, upper) and a hybrid loss balancing point error, interval width, and coverage. It performs well on relatively stationary benchmarks, but its behaviour under OOD financial shift has not been characterised.

We make four contributions:

- 1) The first systematic evaluation of DeepPIPE on NIFTY 50 under a documented price-level shift, showing that single-phase training fails catastrophically (Section IV).
- 2) A two-phase training fix—train the point head alone, then enable interval heads—that restores calibration ($0.32 \rightarrow 0.99$ PICP) without modifying architecture or parameter count.
- 3) DeepPIPE-Ctrl, casting adaptive conformal width control as a contextual bandit with Winkler rewards, 8-dimensional exogenous state, and three policy tiers (Section V).
- 4) Extensive experiments—static and rolling splits, Kupiec/Christoffersen tests, Diebold-Mariano comparisons, VIX adaptiveness, and width-gated backtests—with an honest comparison to conformal PID [5].

II. Related Work

Prediction intervals and scoring rules. Proper scoring rules [2] provide principled training and evaluation targets for probabilistic forecasts. The Winkler interval score is strictly proper: its expectation is minimised uniquely by the correct conditional quantile interval, making it preferable to ad-hoc coverage-width trade-offs.

Deep learning for time series. LSTM, GRU, Transformers [6], [7], and linear baselines [8] dominate point forecasting. Few architectures natively output calibrated

PIs under shift; DeepPIPE is an exception, jointly learning bounds via a hybrid loss.

Conformal and adaptive calibration. Split conformal prediction [3] calibrates residuals post-hoc with finite-sample coverage guarantees. Adaptive Conformal Inference [4] updates the miscoverage level online; conformal PID [5] applies control-theoretic feedback to interval width. These address calibration time but not training-time optimisation pathologies. Our two-phase fix targets the latter; DeepPIPE-Ctrl upgrades static conformal from one global multiplier to a state-conditioned policy.

III. Problem Formulation

Let $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_T\}$ be a multivariate financial series with $\mathbf{x}_t \in \mathbb{R}^d$. Given a look-back window of length L , $\mathbf{X}_t = \{\mathbf{x}_{t-L+1}, \dots, \mathbf{x}_t\}$, we forecast the next-day close y_{t+1} and an interval $[L_{t+1}, U_{t+1}]$ with target coverage p (e.g. 0.90).

Metrics. Coverage (PICP) and mean width (MPIW) are

$$\text{PICP} = \frac{1}{N} \sum_{i=1}^N c_i, \quad \text{MPIW} = \frac{1}{N} \sum_{i=1}^N (U_i - L_i), \quad (1)$$

with $c_i = \mathbb{1}[L_i \leq y_i \leq U_i]$. High PICP with low MPIW is desirable; under shift these objectives conflict.

Distribution shift. Let $P_{\text{train}}(\mathbf{X}, Y) \neq P_{\text{test}}(\mathbf{X}, Y)$. In our study P_{train} spans closes $[7,610, 24,613]$ while P_{test} reaches 26,329, so most test points are OOD in price level. The goal is reliable conditional calibration: intervals should widen when volatility rises and when prices exceed the training envelope.

IV. Two-Phase DeepPIPE Baseline

A. Architecture and hybrid loss

DeepPIPE maps $\mathbf{X}_t$ through an encoder-decoder LSTM to $(\hat{y}, z_\ell, z_u)$; bounds are $L = \hat{y} - \text{softplus}(z_\ell)$ and $U = \hat{y} + \text{softplus}(z_u)$. The hybrid loss is

$$\mathcal{L} = \beta \mathcal{L}_{\text{point}} + \mathcal{L}_{\text{interval}}, \quad \mathcal{L}_{\text{point}} = \text{MAE}(y, \hat{y}), \quad (2)$$

where $\mathcal{L}_{\text{interval}}$ penalises width and miscoverage at $\alpha = 1 - p$.

B. Failure mode under single-phase training

When $\hat{y}$ drifts under shift, simultaneous width minimisation and coverage enforcement produce narrow, miscentred intervals: PICP collapses even though MPIW stays non-zero. Across five seeds we observe mean PICP 0.32 ± 0.27 (worst seed 0.00).

C. Two-phase training procedure

Phase 1: optimise $\mathcal{L}_{\text{point}}$ only (interval heads dormant). Phase 2: activate $\mathcal{L}$ with early stopping on validation MAE. No new parameters are introduced. Mean PICP rises to 0.99 ± 0.01 with improved MAE (894 vs. 1581 index points), consistently across all five seeds (Fig. 1). Fig. 2 shows why: under single-phase training validation

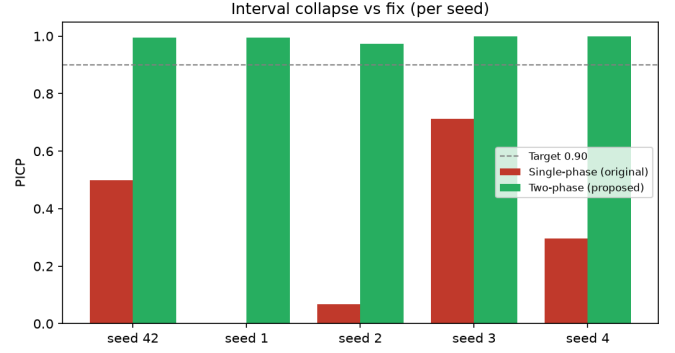


Fig. 1. Per-seed interval coverage. Single-phase training (red) is unstable and frequently far below the 0.90 target; two-phase training (green) is near-perfect across all seeds.

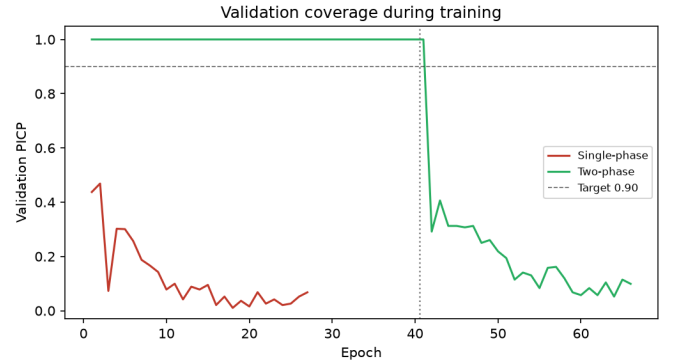


Fig. 2. Validation coverage during training. Single-phase coverage collapses early; the two-phase schedule keeps coverage high through the warm-up (dotted line) and recovers as interval learning proceeds.

coverage decays toward zero, whereas the two-phase schedule holds coverage during the point-only warm-up and re-stabilises after the interval heads activate. Ablations show β dominates λ for calibration (Fig. 3), the warm-up length trades coverage against point error (Fig. 4), and a 14-feature core set generalises better than 21 features with technical indicators.

D. Residual gaps

Two-phase training fixes collapse but not adaptiveness (Table I). Split-conformal calibration applies a single validation-residual quantile $\hat{q}_\alpha$ globally; DeepPIPE-Ctrl generalises this to $m_t \hat{q}_\alpha$ with a learned m_t .

V. DeepPIPE-Ctrl

A. Controller architecture

The two-phase DeepPIPE model and fold-local $\hat{q}_\alpha$ are frozen. At each test time t the controller observes state $\mathbf{s}_t \in \mathbb{R}^8$: VIX level and daily change (train z-scores), realised volatility (5/20-day), momentum (5/20-day), OOD distance $(\text{Close}_t - \max_{\text{train}} \text{Close}) / \max_{\text{train}}$, and rolling

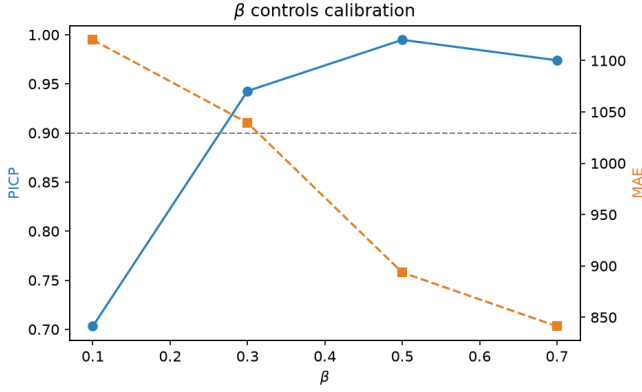


Fig. 3. Effect of the point-prediction weight β . Increasing β raises PICP toward the target and lowers MAE, confirming β (not λ) is the primary calibration lever.

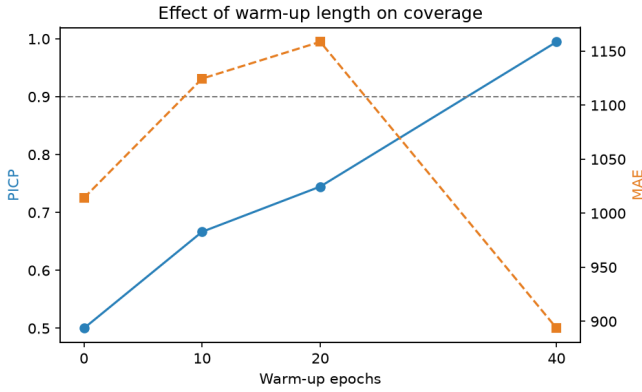


Fig. 4. Warm-up length sweep. Longer point-only warm-up increases coverage; MAE is minimised at the longest warm-up, motivating the two-phase schedule.

mean $|y - \hat{y}|$. All statistics are fold-local (no test leakage). Action a_t selects $m_t \in \mathcal{A} = \{0.5, 0.75, 1.0, 1.25, 1.5\}$ and

$$L_t = \hat{y}_t - m_t \hat{q}_\alpha, \quad U_t = \hat{y}_t + m_t \hat{q}_\alpha. \quad (3)$$

B. Contextual bandit formulation

Because m_t does not affect market dynamics, each step decouples into a contextual bandit with context $\mathbf{s}_t$, action m_t , and reward $r_t = -S(L_t, U_t, y_{t+1})$, where S is the Winkler score at level p :

$$S(L, U, y) = (U - L) + \frac{2}{\alpha}(L - y)\mathbb{I}[y < L] + \frac{2}{\alpha}(y - U)\mathbb{I}[y > U]. \quad (4)$$

Since $\hat{y}_t$, $\hat{q}_\alpha$, and y_{t+1} are known retrospectively, the full reward matrix $\mathbf{R} \in \mathbb{R}^{N \times |\mathcal{A}|}$ is observable offline—no importance-weighted off-policy estimation is required in v1.

C. Policies and baselines

T0 (GreedyConstant): single best constant multiplier. T1: LinUCB / LinTS linear bandits on $\phi(\mathbf{s}_t)$. T2: MLP

TABLE I
Residual calibration gaps motivating DeepPIPE-Ctrl.

Gap	Phenomenon	Evidence
G1	Volatility-blind widths	$\rho(\text{width}, \text{VIX}) = -0.50$
G2	Regime-break under-coverage	Fold 1 PICP = 0.01
G3	Over-coverage, level-blind	PICP ≈ 0.995

softmax policy trained by reward regression. Static conformal: $m_t \equiv 1$. Conformal PID: feedback on coverage error with clipped multipliers [5].

D. Evaluation objectives

Inner (statistical): PICP, MPIW, mean normalised interval score, Kupiec POF, Christoffersen independence, $\rho(\text{width}, \text{VIX})$. Outer (economic): a width-gated long-only backtest where position size is inversely proportional to interval width; we report Sharpe and max drawdown. P&L is not optimised during training; it is a downstream consequence of calibration quality.

VI. Experimental Setup

Data. Daily NIFTY 50, 1,676 trading days (2020–2026), 14 core features (OHLCV, India VIX, global indices, commodities, FX). Chronological split: 70% train / 15% val / 15% test ($n_{\text{test}} = 192$).

Training. $L=60$, hidden 64, 2 layers, dropout 0.2, $\beta=0.5$, $\lambda=15$, $p=0.90$, Adam lr 5×10^{-4} , Phase 1: 40 epochs, Phase 2: 120 epochs (early stop).

Protocols. E1–E10 cover the main comparison, fail/fix seeds, β/λ sweeps, feature ablation, robustness, rolling-window (5 folds), calibration tests, DM/bootstrap, VIX analysis, conformal baseline, and training dynamics. E11–E14 cover controller inner metrics, VIX–width, a rolling stress test, and the width-gated backtest. Experiments run in the open-source deeppipe package (Python 3.10+, PyTorch 2.x) with YAML configuration and a walk-forward harness.

VII. Results

A. Two-phase baseline (E1–E2)

Table II summarises the main test-set comparison. DeepPIPE two-phase achieves PICP 0.995 but MAE 893.6, higher than naive baselines under OOD shift—consistent with prioritising calibration over point accuracy. Table III shows the fail/fix contrast across five seeds. Fig. 5 visualises the resulting point forecast and 90% interval over the test period: the band expands during the sharp late-window drawdown, keeping the realised close inside the interval.

B. Volatility adaptiveness (E8)

Two-phase DeepPIPE exhibits $\rho(\text{width}, \text{VIX}) = -0.50$; mean widths across low/mid/high VIX terciles are nearly identical (3861/3838/3824), confirming gap G1.

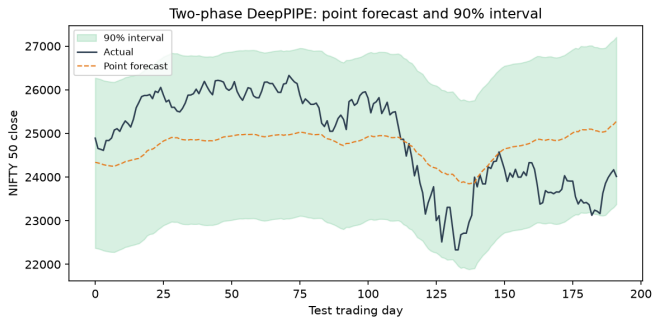


Fig. 5. Two-phase DeepPIPE point forecast and 90% prediction interval over the test period. The interval contains nearly all realised closes, including the volatile drawdown near day 130.

TABLE II
Main comparison on held-out test set (E1).

Model	MAE	RMSE	PICP	MPIW
DeepPIPE (two-phase)	893.6	984.8	0.995	3841.1
Naive	151.6	209.8	—	—
Moving average (5)	235.5	302.9	—	—
Linear regression	418.6	540.8	—	—
GRU	458.7	615.3	—	—
LSTM	2016.8	2155.7	—	—

C. DeepPIPE-Ctrl inner metrics (E11)

Table IV compares controllers on the static test split ($p=0.90$). Conformal PID is closest to target coverage and reverses the VIX-width sign. T0 matches static conformal exactly, so no constant $m \neq 1$ helps here. Learned bandits underperform PID on interval score; T2 severely undercovers.

D. Rolling-window stress test (E13)

On Fold 1 (prices 21.8k $\rightarrow$ 26.0k), LinUCB PICP improves from 0.011 (static) to 0.033—directionally better but far from 0.90. On stable folds, LinUCB matches or marginally improves normalised interval score. Fig. 6 makes the mechanism explicit: coverage collapses precisely on the fold whose maximum price exceeds the training maximum (24,613), confirming gap G2.

E. Width-gated backtest (E14)

Table V reports the outer economic metrics. All Sharpe ratios are negative on this OOD window; PID is least negative, aligning with its superior inner calibration.

VIII. Discussion

Training vs. calibration. Results support a two-layer view: two-phase training repairs optimisation-induced interval collapse, while adaptive conformal methods address post-hoc miscalibration. They are complementary; static split-conformal after two-phase training still over-covers and ignores volatility state.

Learned vs. hand-crafted control. With only ~ 190 validation and ~ 192 test points per fold, linear bandits

TABLE III
Single-phase vs. two-phase training (five seeds).

Metric	Single-phase	Two-phase
PICP (mean $\pm$ std)	0.32 ± 0.27	0.99 ± 0.01
MAE (mean)	1580.9	894.0
MPIW (mean)	1226.9	4121.8

TABLE IV
Adaptive calibration methods ($n=192$, target PICP 0.90).

Method	PICP	MPIW	Score	$\rho_{w,VIX}$
Static conformal / T0	0.885	2694.5	2.39	—
Conformal PID	0.901	3193.9	2.54	0.020
LinUCB (T1)	0.865	2820.3	2.68	-0.42
LinTS (T1)	0.844	2810.9	2.67	-0.36
MLP (T2)	0.719	2552.9	3.30	-0.58

and MLPs lack data to beat a well-tuned PID controller. This is an honest negative result: adaptive width need not require learning if a simple feedback law suffices. Future work should add coverage-error memory (a true MDP), asymmetric bounds, and online updates (ACI/CQL).

Limitations. The controller rescales width around a possibly biased $\hat{y}$; it does not correct location error. The economic evaluation is illustrative, not a trading objective. Results cover one index and one shift pattern; broader asset panels would strengthen external validity.

IX. Conclusion

We showed that single-phase DeepPIPE training fails under a documented NIFTY 50 distribution shift while two-phase training restores coverage. We then introduced DeepPIPE-Ctrl, formulating adaptive conformal width control as a Winkler-driven contextual bandit with exogenous market state. Conformal PID currently dominates learned policies on calibration metrics; bandit policies show modest gains only under extreme rolling regimes. Together, two-phase training plus adaptive calibration form a practical pipeline for uncertainty-aware forecasting under shift, with open-source code for full reproduction.

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Author Contributions

The overall research direction—problem framing (gaps G1–G3), the contextual-bandit formulation, and the evaluation protocol—was designed jointly by all three authors. Individual ownership of the delivered work was as follows.

Raunak Goyal (Layer 1: two-phase DeepPIPE). Led the base forecaster: the encoder-decoder DeepPIPE model, the two-phase training schedule, the data pipeline and feature engineering, and the statistical validation suite

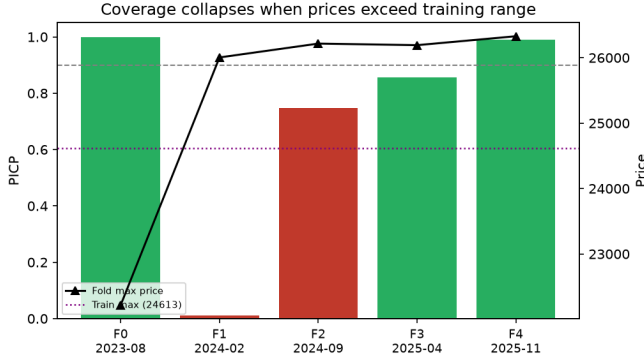


Fig. 6. Walk-forward coverage per fold (bars) against each fold’s maximum price (line). Coverage collapses on the fold that first breaches the training price ceiling.

TABLE V
Outer economic metrics, width-gated strategy.

Method	Ann. return	Sharpe	Max DD
Static conformal / T0	−3.8%	−0.27	−15.2%
Conformal PID	−1.6%	−0.12	−14.1%
LinUCB	−7.3%	−0.58	−14.6%
MLP	−4.8%	−0.36	−14.8%

(Diebold–Mariano, bootstrap CIs, Kupiec and Christofersen tests). Ran experiments E1–E10 and produced Figs. 1–5.

Anshul Saxena (Layer 2: DeepPIPE-Ctrl). Led the adaptive controller: the 8-dimensional state features, the Winkler reward and offline bandit dataset, the T0/T1/T2 policies (GreedyConstant, LinUCB/LinTS, MLP), the rollout loop, and the width-gated economic backtest. Ran experiments E11–E14 and produced Fig. 6 and the controller results.

Snehanshu Baneerjee (systems, baselines, and reproducibility). Led the software engineering: the installable deeppipe package, typed YAML configuration, the Typer CLI, the pipeline/artifact layer, the unit and integration test suite, and the CI workflow. Implemented the static-conformal and conformal-PID baselines and owned experiment reproducibility.

All authors contributed to analysis, manuscript writing, and cross-review before integration.

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